



Habitat Classifier 1.0 Handbook

System Requirements

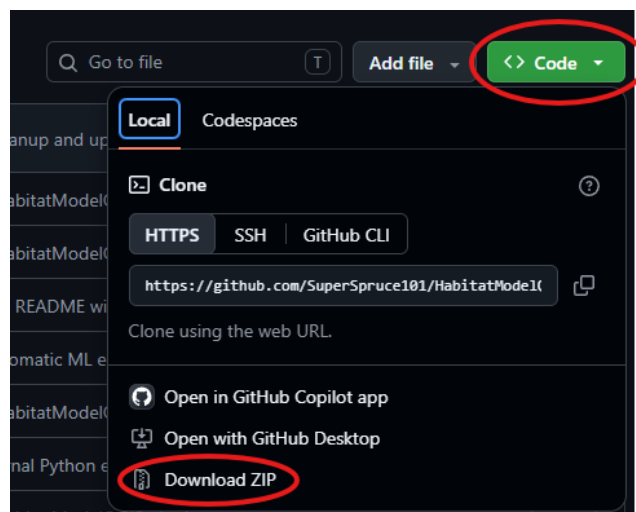
The Habitat Classifier plugin is designed for use with QGIS 3.x and uses a dedicated Python 3.11 environment for machine-learning inference and geospatial processing.

requirements:

- QGIS 3.x
- Python 3.11
- Internet connection during first-time setup
- NVIDIA GPU recommended for faster processing
- Sufficient storage for the Python environment, model files, and raster outputs

Installation

1. Download the plugin ZIP file from <https://github.com/SuperSpruce101/HabitatModelQGIS>



2. Download Python 3.11 from <https://www.python.org/ftp/python/3.11.8/python-3.11.8-amd64.exe>

3. In QGIS, open:

Plugins > Manage and Install Plugins > Install from ZIP

Select: “*HabitatModelQGIS.zip*”

Once installed, the Habitat Classifier icon should appear in the QGIS toolbar, and the tool will also be available through the Processing Toolbox.

First-Time Environment Setup

On first use, the plugin automatically prepares the Python 3.11 environment required to run the classification model. The environment contains the packages needed.

4. In QGIS open:

Plugins > Habitat Classifier > Habitat Classifier Settings

5. The plugin should identify if hardware supports NVIDIA GPU acceleration

6. Select setup environment (this may take a while but only needs to be done once)

7. One complete, test and save the python environment.

Usage Instructions

Select the Habitat Classifier logo from the task bar or toolbox.

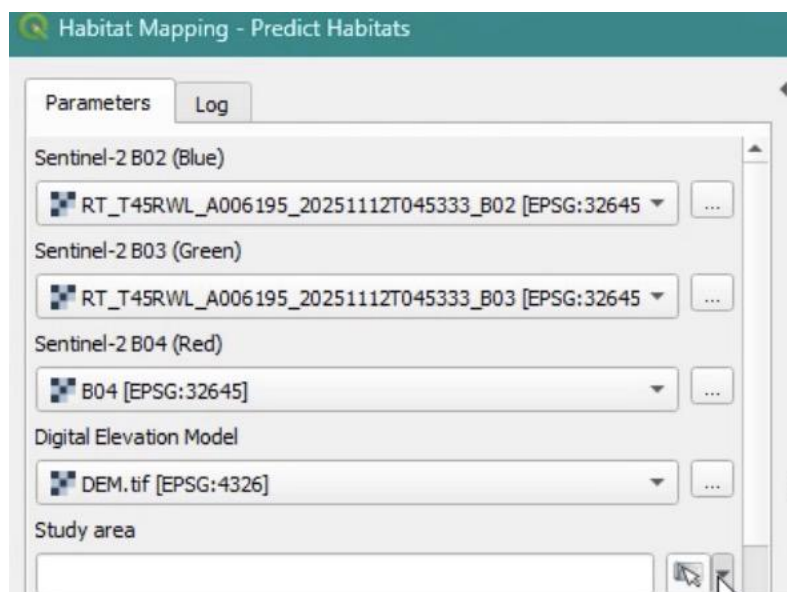
1. Input Parameters:

Three separate Sentinel 2 RGB bands (these should appear in greyscale) the plugin accepts RT (reflectance transformed) and Raw .jp2 files

DEM Global 30m (recommend using open-topography plugin from QGIS repository)

2. Select Area:

User can then define the area for habitat prediction to generate using the “study area” option, this can then be selected from a layer or drawn onto the map canvas.



3. Predict Habitats

Further Output Options

Beyond the primary habitat prediction map, the plugin generates several additional geomorphological outputs that provide further environmental context, these are **toggled ON by default**. Predicted forest areas can be subdivided into Gentle/Plateau Forest, Steep-Slope Forest and Valley Forest using DEM-derived slope and Topographic Position Index (TPI). Flow accumulation is also used to identify likely drainage pathways and potential river networks, while areas of concentrated flow can be used to refine Bare Terrain predictions. Together, these outputs extend the model beyond broad land-cover classification by incorporating information about the terrain processes and landscape position associated with different habitat types.

The plugin also outputs the cropped DEM and RGB bands, this can also be toggled off before running “*Predict Habitats*”.